

JEE MAIN REVISION MASTER: CHEMICAL EQUILIBRIUM 2026 2 QUESTIONS 3.33% WEIGHTAGE ↓33.4%

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Equilibrium Constants (K_p , K_c) & Reaction Quotient	8 PYQs	Equilibrium Constants
Moderate	Le Chatelier's Principle & Stress Factors	3 PYQs	Le Chatelier

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Equilibrium Constants (K_p , K_c) & Reaction Quotient	$K_p = K_c(RT)^{\Delta n_g}$, $\Delta G^\circ = -RT \ln K = -2.303RT \log_{10} K$, $Q < K \implies \text{Forward}$
Le Chatelier's Principle & Stress Factors	$\ln \left(\frac{K_2}{K_1} \right) = \frac{\Delta H^\circ}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$ (van 't Hoff)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Equilibrium Constants (K_p , K_c) & Reaction Quotient (High Yield — 8 PYQs)

Core Principle: Relationship between K_p and K_c , expression of equilibrium constants, and spontaneous direction based on Q vs K .

■ **Caution / JEE Pitfall:** In calculating Δn_g , count ONLY gaseous species: $\Delta n_g = \sum n_{\text{(gas)}}(\text{products}) - \sum n_{\text{(gas)}}(\text{reactants})$. Pure solids and liquids are excluded ($a = 1$).

■ **Speed Trick:** If a reaction is multiplied by factor n , its new equilibrium constant becomes $K' = K^n$. If reversed, $K' = 1/K$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "For the reversible reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ When the partial pressure is measured i..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "The equilibrium constant at 298 K for the reaction $A + B \rightleftharpoons C + D$ is 100. If the initial concentrations of all the four speci..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Which of the following statements is/are true about equilibrium? (a) Equilibrium is possible only in a closed system of at a given temperatu..." [Ans: N/A]

3.2 Le Chatelier's Principle & Stress Factors (Moderate Yield — 3 PYQs)

Core Principle: Dynamic response of equilibrium systems to variations in concentration, temperature, pressure, and inert gas addition.

■ **Caution / JEE Pitfall:** Addition of an inert gas at CONSTANT VOLUME has NO effect on equilibrium position because partial pressures remain unchanged!

■ **Speed Trick:** Inert gas addition at CONSTANT PRESSURE shifts equilibrium toward the side with MORE gaseous moles ($\Delta n_g > 0$ shifts forward).

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "For the following gaseous reversible reaction: $3\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{A}_{(3)}\text{B}_{(g)}$ ($\Delta_r H = -q$)..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "According to Le Chatelier's principle, in the reaction $\text{CO}(g) + 3\text{H}_2(g) \rightleftharpoons \text{CH}_4(g) + \text{H}_2\text{O}(g)$, the formation of methane i..." **[Ans: N/A]**
- **[2018 Shift] Q#1 (MCQ):** "The relationship between K_p and K_c is $K_p = K_c(RT)^{\Delta n}$. What would be the value of Δn for the reaction $\text{NH}_4\text{Cl} \rightleftharpoons \text{NH}_3 + \text{HCl}$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

For the following gaseous reversible reaction: $3\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{A}_{(3)}\text{B}_{(g)}$ ($\Delta_r H = -q$ kJ). The amount of product $\text{A}_{(3)}\text{B}_{(g)}$ is affected by ____

- (A) Temperature alone
- (B) Pressure alone
- (C) Both temperature and pressure
- (D) Temperature, pressure and catalyst

Correct Answer: None

Solution Analysis: For the reaction $3\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{A}_{(3)}\text{B}_{(g)}$ the forward reaction is exothermic because $\Delta_r H = -q$ kJ. So, heat is released in the forward direction. If temperature is increased, the equilibrium shifts in the backward direction to absorb heat. So the amount of $\text{A}_{(3)}\text{B}$ decreases. If temperature is decreased, the equilibrium shifts in the forward direction. So the amount of $\text{A}_{(3)}\text{B}$ increases. Now look at pressure. On the reactant side, the number of gaseous moles is $3+1=4$. On the product side, the number

Benchmark Problem 2: KCET 2026 (Q#2)

For the reversible reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ When the partial pressure is measured in atmosphere, the value of K_p at 500°C is 1.44×10^{-5} . The value of K_c when the concentration is expressed in mol L^{-1} is:

- (A) $\frac{1.44 \times 10^{-5}}{(0.082 \times 500)^{-2}}$
- (B) $\frac{1.44 \times 10^{-5}}{(8.314 \times 773)^{-2}}$
- (C) $\frac{1.44 \times 10^{-5}}{(0.082 \times 773)^2}$
- (D) $\frac{1.44 \times 10^{-5}}{(0.082 \times 773)^{-2}}$

Correct Answer: None

Solution Analysis: For the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ we use the relation between K_p and K_c :

$K_p = K_c(RT)^{\Delta n}$ where $\Delta n = \text{moles of gaseous products} - \text{moles of gaseous reactants}$ Here, $\Delta n = 2 - (1 + 3) = 2 - 4 = -2$ So, $K_p = K_c(RT)^{-2}$ Hence, $K_c = \frac{K_p}{(RT)^{-2}}$ Given temperature is 500°C , so we convert it to kelvin:

$T = 500 + 273 = 773\text{K}$ Since pressure is in atmosphere and concentration is in mol L^{-1} , we use $R = 0.082\text{L atm mol}^{-1}\text{K}^{-1}$

Thus, $K_c =$